1 Planning (15 marks)

Defining the problem (3 marks)

P1 *A* is the independent variable or vary *A*. [1]

P2 *R* is the dependent variable or determine *R* for different *A*. [1]

P3 Keep the temperature (of glass) constant. Do not allow "controlled variable". [1]

Methods of data collection (5 marks)

M1 Basic circuit diagram. [1]
Ammeter and voltmeter with power supply, or
ohmmeter without power supply, or
bridge methods.

M2 Correct orientation of glass between electrodes – largest cross-sectional area. [1]

M3 A distance (thickness) measured using a micrometer/vernier scale/vernier callipers. [1]

M4 Method of determining area perpendicular to current flow. [1]
Distances measured and multiplied together.
This mark may only be scored if it is clear that the correct dimensions are being used.

M5 Method of determining resistance. [1]
Ohmmeter.
$R = V/I$ justified.
Description of balancing bridge with correct equation.

Method of analysis (2 marks)

A1	A2
R against $1/A$	ρ = gradient/l
R against l/A	ρ = gradient
$1/A$ against R or $1/R$ against A	$\rho = 1/(\text{gradient} \times l)$
l/A against R or l/R against A	ρ = 1/gradient
lg R against lg A	$\rho = 10^{l \times \text{y-intercept}}$

[2]

Safety considerations (1 mark)

S1 Relevant safety precaution related to: [1]
EHT power supply (>100 V) – switch off before changing circuit/use of rubber gloves;
or handling glass – wear (thick) gloves.

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]
Calculation of typical resistance of glass using value of resistivity given.
Range of ammeter or ohmmeter with reasoning.
Use of EHT or power supply >1000 V or microammeter/galvanometer.
Take many readings of thickness and average.
Good contact between circuit and glass e.g. metal plates, foil, conducting putty.
Metal plates/foil/conducting putty to cover all of the cross-sectional area in use.
Method of securing good contact between circuit and glass, e.g. g clamps, weights.
Clean/dry the glass.

[Total: 15]

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